

VIJAY VARDHAN REDDY YAMMANURU

Data Analyst | Business Analyst | AI/ML Developer | B.Tech Big Data, NIIT University
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PROFESSIONAL SUMMARY

Results-driven 3rd-year B.Tech (Big Data) student at NIIT University with a proven track record of building production-grade, data-intensive systems. Engineered a real-time F1 Prescriptive Analytics engine using Monte Carlo Tree Search achieving 89.1% strategy alignment against live 2026 race telemetry. Built NeuroLens, a full-stack AI application powered by Meta AI's TRIBE v2 model that predicts student cognitive load across 20,484 brain vertices. Core strengths in SQL, Power BI, Python (FastAPI, Pandas), React.js, Redis, and end-to-end data pipeline architecture. Certified by Google, Deloitte (Forage), and TATA (Forage). Demonstrated leadership managing 100+ volunteers at NIIT University's flagship tech fest. Seeking a Data Analyst or Business Analyst internship to deliver measurable impact.

TECHNICAL SKILLS

Data & BI	Microsoft Power BI (DAX, Power Query, Import Mode) · MySQL · Microsoft Excel · Pandas
AI / ML	Monte Carlo Tree Search (MCTS) · Meta AI TRIBE v2 · FastF1 API · scikit-learn · NumPy
Backend	Python 3.10+ (FastAPI, concurrent.futures) · Node.js · Express.js · Redis Pub/Sub · WebSocket
Frontend	React 18 · Three.js (3D / WebGL) · Vite · JavaScript
Databases	MySQL · MongoDB (Mongoose ODM) · Redis (in-memory)
Cloud & Infra	Kaggle P100 GPU · ngrok · HuggingFace Hub · Vercel · Render · Git
Soft Skills	Team Leadership · Stakeholder Management · Cross-functional Coordination

PROJECTS

F1 Real-Time Prescriptive Analytics Engine | Python · MCTS · Redis · React · FastF1 API 2026

- Engineered a deterministic Prescriptive AI that simulates 8,000+ alternate race futures per lap using Monte Carlo Tree Search (MCTS) + Deep-Branching Heuristics — calculating the mathematically optimal pit-stop window live during a race.
- Solved Python GIL bottleneck by engineering a multi-threaded compute pipeline using concurrent.futures.ThreadPoolExecutor, achieving < 0.15 seconds decision time per driver — processing all 20 cars before reaching the next corner.
- Built a dual-pipe Redis Pub/Sub microservices architecture: Pipe A streams raw FIA telemetry (via FastF1 API) to the MCTS engine; Pipe B injects enriched strategic commands (PIT → HARD / STAY OUT) to a React.js live dashboard.
- Back-tested against live 2026 Australian, Chinese, and Japanese Grand Prix telemetry — achieved 89.1% Global Strategy Alignment. Outperformed Mercedes pit wall by 2 laps at Suzuka (Lap 35 vs Lap 37), projecting a 1.5-second race-time saving.
- Engineered full fault-tolerance: system autonomously handles null telemetry, detected a DNS at Australia GP, instantly sanitized dirty data, updated the grid state to 19 cars, and continued simulating without dropping a frame.

IPL Analytics Dashboard | SQL · MySQL · Power BI 2024

- Collected, cleaned, and analyzed a 200,000+ row IPL dataset (2008–2023) covering match results, player performance, and venue data using MySQL as the primary data warehouse.
- Engineered complex SQL logic (CASE WHEN, CREATE TABLE, WHERE, ORDER BY) to pre-calculate Winner, Runner-Up, Orange Cap, and Purple Cap metrics — offloading heavy computation from Power BI to MySQL, dramatically improving dashboard performance.
- Built an interactive neon-themed Power BI dashboard connected directly to MySQL via Import Mode; a master Season slicer drives dynamic KPI cards, Points Table bar chart, Toss Analysis pie chart, and Venue Analysis column chart.

NeuroLens — Cognitive Load Analyzer for Educators | React · FastAPI · Three.js · Meta AI TRIBE v2 · Kaggle P100 GPU 2026

- Built a full-stack AI web app that predicts second-by-second student brain activation across 20,484 cortical vertices using Meta AI's TRIBE v2 foundation model (peer-reviewed research, Benchetrit et al., 2025 — 2–3× better than prior methods).
- Architected a tri-modal AI inference pipeline: V-JEPA2 (video frames) + wav2vec 2.0 (audio) + Llama 3.2 (transcript) fused by a Transformer into unified brain representations — deployed on Kaggle P100 GPU (16GB VRAM), tunneled via ngrok.
- Built a custom Three.js 3D brain viewer from scratch — wrote a binary GLB parser without external dependencies, mapped TRIBE v2 activations as per-vertex fMRI heatmap colors across 6 mesh parts at 60fps with smooth lerp animation.
- Engineered a neuroscience-based alert detection system: identifies Severe Overload (PFC > 0.70, Attention < 0.40 for ≥3s), Peak Engagement (Attention > 0.72 for ≥10s) — gives educators exact timestamps like '4:32–6:18: students disengaged.'
- Designed a fully async FastAPI backend with non-blocking GPU inference job queue; implemented real-time progress streaming and built the entire UI with zero CSS frameworks — pure inline styles matching Meta AI TRIBE v2 research site.

Hoshitha Boutique — Full-Stack E-Commerce | MERN Stack (React · Node.js · MongoDB)

2024–25

- Architected and deployed a decoupled MERN application: React.js + Vite on Vercel, Node.js/Express.js on Render, MongoDB for e-commerce data — with bcryptjs authentication, RBAC, and automated Nodemailer/SMTP order notifications.
- Diagnosed CORS conflicts between Vercel and Render; resolved IPv6 timeout issues by routing SMTP through Port 587. Improved SEO via Open Graph tags and Google Search Console indexing — live and publicly accessible.

LEADERSHIP & EXTRA-CURRICULARS

Operations Team Core & Lead | siNUoid v9 — NIIT University Annual Tech Fest

2024-26

- Held the highest Operations position — directed 6 team leads and 100+ volunteers across a 3-day event hosting 5,000+ attendees. Managed full operational accountability: backstage, DJ night, vendor negotiations, inventory, and crowd control across 20+ events.
- Spearheaded complete LAN/Wi-Fi infrastructure deployment and live tech support for the gaming event arena. Awarded Certificate of Appreciation + Letter of Recommendation by Maj. Gen. A K Singh, Dean of Student Affairs, NIIT University.
- Managed a dedicated team of volunteers coordinating logistics, event curation, and technical support across multiple event verticals. Awarded Certificate of Appreciation by Maj. Gen. Ashok K Singh (Retd.), Dean of Student Affairs, NIIT University.

Head of Elections | NIIT University Student Council Elections

2024

- Organized the complete university election lifecycle — candidate nominations, campaign regulation, ballot box management, vote counting, and official result declaration; managed crowd control and conflict resolution campus-wide.

EDUCATION

NIIT University, Neemrana, Rajasthan

2023 – 2027

Bachelor of Technology in Big Data

CERTIFICATIONS & ACHIEVEMENTS

- Data Analytics Job Simulation (Data Analysis · Forensic Technology) — Deloitte via Forage

Dec 2025
- Data Visualisation: Empowering Business with Effective Insights — TATA via Forage

Dec 2025
- Google AI Essentials — Google Career Certificates (Credly Verified)

Dec 2025
- SQL Essential Training (7.8 CPE Credits — NASBA Accredited) — LinkedIn Learning

Feb 2026
- Complete Your First Project in SQL — LinkedIn Learning

Feb 2026
- Operations Core Certificate + Letter of Recommendation from Dean — NIIT University — siNUoid v9

Nov 2025